

Maxillofacial Trauma

Guideline Title: Maxillofacial Trauma

Guideline Number: GUID-3203-T007

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- I. **PURPOSE:** To recognize the patient with maxillofacial trauma and to provide adequate airway control. Provide adequate pain management and transport the patient to a facility with the ability to definitively care for this type of patient.
- II. **PATHOPHYSIOLOGY:**
 - a. There are three major types of Le Fort fractures:
 - i. **Le Fort I - transverse maxillary fractures:** The most common maxillary fracture. It is a fracture of the transverse alveolar process at the nasal floor. It extends from the upper teeth into the nose. Signs and symptoms include malocclusion of teeth/anterior open bite, moveable maxilla, and/or epistaxis.
 - ii. **Le Fort II - pyramid fractures:** A Le Fort II fracture is a pyramidal fracture that extends through the central part of the maxilla up through the superior nasal area. It also usually includes the orbits. Signs and symptoms include the nose and dental arch moving together when manipulated, epistaxis, periorbital edema and ecchymosis, subconjunctival hemorrhage, infraorbital paresthesia, and/or possible extraocular muscle entrapment.
 - iii. **Le Fort III - complete or "free floating face":** When this fracture occurs, a total craniofacial separation result. It usually also involves the orbits. Signs and symptoms: massive bleeding, usually loss of consciousness, facial bones move without frontal bone movement when manipulated, CSF rhinorrhea, epistaxis, periorbital edema and ecchymosis, subconjunctival hemorrhage, extraocular muscle entrapment, and/or infraorbital paresthesia.
 - b. Up to 50% of adult and 70% of pediatric patients with maxillofacial injuries will show associated injuries of the cranium, cervical spine, orbit, chest, or abdomen.
- III. **DEFINITIONS:**
- IV. **DEPARTMENT GUIDELINE:**
 - a. Establish care as defined by *GUID-G002 General Patient Care* and *GUID-T008 Multi-system Trauma*.
 - b. Ensure ability to maintain an adequate and effective airway, and/or definitive airway while maintaining C-spine precautions, full head, and spinal immobilization.
 - c. Provide supplemental high flow oxygen.
 - d. Unless there is evidence of airway compromise, dislodged teeth should left in place or transported in a container of sterile saline.

- e. Patients with severe oropharyngeal bleeding who do not require spinal immobilization can assume a position of comfort by leaning forward and suction themselves.
- f. If spinal immobilization is required, patient is alert, and bleeding is not severe, the backboard may be tilted with the bleeding side down, to protect the airway and allow suctioning.
- g. Intubation may be necessary to protect the airway per *GUID-G008 Rapid Sequence Induction for Endotracheal Intubation* and *GUID-PR020 Endotracheal Intubation* or *GUID-PR007 Cricothyrotomy – Surgical*. Paralytic medications should be avoided.
 - i. Removal of debris and aggressive suctioning with two suction devices may be necessary.
 - ii. Anticipate difficulty in patients with heavy bleeding, evidence of a hematoma extending into the neck, or supraclavicular region.
 - iii. Simultaneous preparation for oral intubation & cricothyrotomy (i.e., double set-up) is prudent.
 - iv. On rare occasions, vascular injuries in the face cause significant hematomas that extend into the neck or supraclavicular region with little external bleeding. Anatomic distortion of the pharynx & neck can make intubation and cricothyroidotomy difficult. In such circumstances, if transport time is short, definitive airway management maybe best deferred until the patient reaches the emergency department provided the following conditions are met:
 1. Patient is conscious, protecting their airway, and breathing adequately with SpO2 maintained >90% -OR- Ventilation can be performed without difficulty using a bag-mask with SpO2 > 90%
 2. Flight team should notify the receiving institution of the patient's condition so preparations can be made for a difficult intubation.
- h. Following intubation, large amounts of gauze can be placed in the oropharynx and nasopharynx to obtain hemorrhage control via direct pressure.
- i. Control hemorrhage with well-aimed direct pressure. If unable to control with direct pressure, consider *GUID-PR028 Hemostatic Dressing - QuickClot Gauze*.
- j. Ensure adequate IV/IO access.
- k. Avulsed tissue, including ears, the tip of the nose, or large skin flaps, should be transported with the patient in saline-soaked gauze.
- l. Assess for rhinorrhea, which may represent a cerebrospinal fluid (CSF) leak
- m. Treat pain and/or anxiety per *GUID-G005 - Analgesia and Sedation Management*.
- n. Treat nausea/vomiting per *GUID-G004 – Motion Sickness/Nausea*.
- o. Closely monitor vital signs.
- p. Consider TXA administration per *GUID-3203-T016 - Tranexamic Acid for Severe Hemorrhage*.
- q. Re-assess airway patency, neurological status, vital signs, and bleeding frequently.

V. **DOCUMENTATION:** EMS Charts

VI. **REFERENCES:**

- a. Leonard J, Zietlow J, Morris D, Berns K, Eyer S, Martinson K, et al. (2016). A multi-institutional study of hemostatic gauze and tourniquets in rural civilian trauma. *J Trauma Acute Care Surg*, 81(3), 441–4.
- b. Mayersak, R. J., Moreira, M. E., & Grayzel, J. (2021). Initial evaluation and management of facial trauma in adults. *UpToDate*. Last updated Feb 6, 2020.